Exam 1 Review Solutions – PSTAT 5LS

****Multiple Choice** Please circle the best answer.**

1. We randomize in an experiment because
    a. we want the volunteers to be as representative of the population as possible.
    b. the treatment groups will be exactly the same.
    c. randomization allows us to equalize the effects of unknown/uncontrollable sources of variation.
    d. we need to identify groups that might respond differently to the treatments.

2. If we know that a deviation from the mean for one data point is 0, then we know that the:
    a. distribution is symmetric.
    b. the observation is equal to the mean.
    c. standard deviation is also 0.
    d. median would be a more accurate representation of center than would the mean.

3. To study the effectiveness of a zinc nasal spray for reducing the duration of a common cold, researchers recruited 104 subjects who agreed to report to their lab within 24 hours of getting cold symptoms. Each subject was randomly assigned to one of three groups: one received full dosage of the zinc spray, another received a low dosage, and a third received a placebo spray. The cold symptoms lasted an average of 1.5 days for the full dosage group, 3.5 days for the low dosage group, and 10 days for the placebo group. Which of the following is most appropriate for this experiment?
    a. Explanatory variable: length of time cold symptoms lasted
    Response variable: amount of zinc spray given (full, low, placebo)
    b. Explanatory variable: amount of zinc spray given (full, low, placebo)
    Response variable: length of time cold symptoms lasted
    c. Explanatory variable: severity of the cold
    Response variable: amount of zinc spray given (full, low, placebo)
    d. Explanatory variable: severity of the cold
    Response variable: length of time cold symptoms lasted

4. If a distribution is clearly skewed to the right,
    a. the mean is less than the median.
    b. the mean is about the same as the median.
    c. the mean is greater than the median.

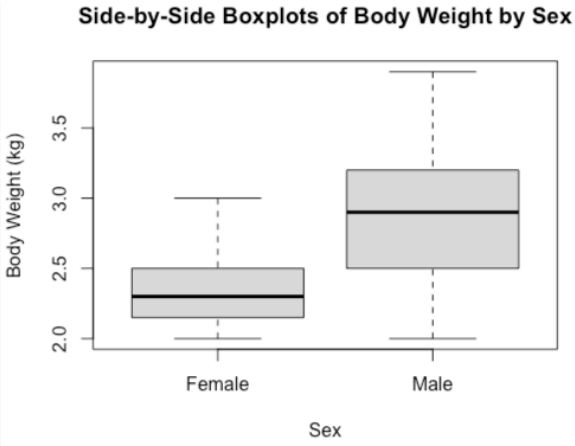
5. Vivek has an average bowling score of 180 and bowls in a league where the average for all bowlers is 150 and the standard deviation is 20. Varsha has an average bowling score of 190 and bowls in a league where the average is 160 and the standard deviation is 15. Who ranks higher in the own league, Vivek or Varsha?
    a. Varsha because her 190 is higher than Vivek's 180.
    b. Varsha because her z-score is higher than Vivek's.
    c. Vivek because his z-score is higher than Varsha's.
    d. Varsha and Vivek have the same rank in their leagues because both are 30 pins above the mean in their league.

6. The Centers for Disease Control and Prevention (CDC) administers a large number of survey programs for monitoring the status of health and health care in the United States. One of these programs is the National Hospital Ambulatory Medical Care Survey (NHAMCS), which includes information from hospital records for a random sample of individuals treated in hospital emergency rooms around the country. To what population can we reasonably generalize findings from the NHAMCS?
   a. The random sample of hospital emergency rooms that were surveyed
   b. Individuals treated in hospital emergency rooms in the United States
   c. All adults in the United States

7. Which of the following statements best describes the relationship between a parameter and a statistic?
   a. A parameter is used to estimate a statistic.
   b. A statistic is used to estimate a parameter.
   c. The value of a statistic will always be smaller than the value of a parameter.
   d. The value of a parameter is always known, but the value of a statistic is rarely ever known.

8. Four hundred (400) student volunteers were recruited from a university's recreation center to participate in a study. First, everyone took an anxiety test. Then, half of the participants were randomly assigned to participate in a pet therapy stress-relief intervention, in which they pet dogs and cats for an hour. The other half of the participants were in a control group that received an hour-long lecture on stress-relief tips. Later, everyone was given the anxiety test again, and it was found that the pet therapy group had a significantly greater reduction in anxiety than the control group (p-value < 0.001). Can we conclude that the pet therapy intervention likely caused this significantly greater reduction in anxiety?
   a. No, because the students were not randomly assigned to treatments.
   b. No, because the students who volunteered may not be representative of all students at the university.
   c. Yes, because the students who volunteered will be representative of all students at the university.
   d. Yes, because the students were randomly assigned to the treatment groups.

Use the following information to answer Questions 9-12:
A researcher was interested in studying the relationship between the heart weight and total body weight in domestic cats. For each of 144 cats, the researcher recorded the sex, body weight (kg), and heart weight (g) of the cat. The following is a side-by-side boxplot of the body weights of the cats by sex.



Use the boxplots to make quantitative comparisons. In each case, select the most appropriate statement from the following choices (you may use each choice more than once or not at all):

- Quantity A is greater
- Quantity B is greater
- The quantities are about the same
- The relationship cannot be determined without more information

9. **Quantity A:** The maximum body weight for female cats
   Quantity B: The maximum body weight for male cats
   a. Quantity A is greater
   b. Quantity B is greater
   c. The quantities are about the same
   d. The relationship cannot be determined without more information

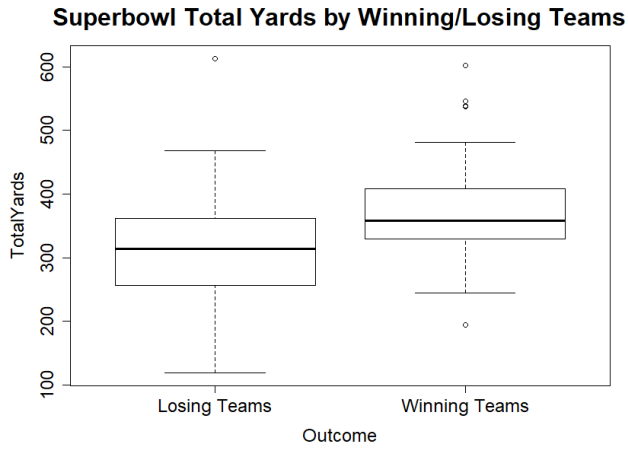
10. **Quantity A:** The 75th percentile of body weights for female cats
    Quantity B: The 25th percentile of body weights for male cats
    a. Quantity A is greater
    b. Quantity B is greater
    c. The quantities are about the same
    d. The relationship cannot be determined without more information

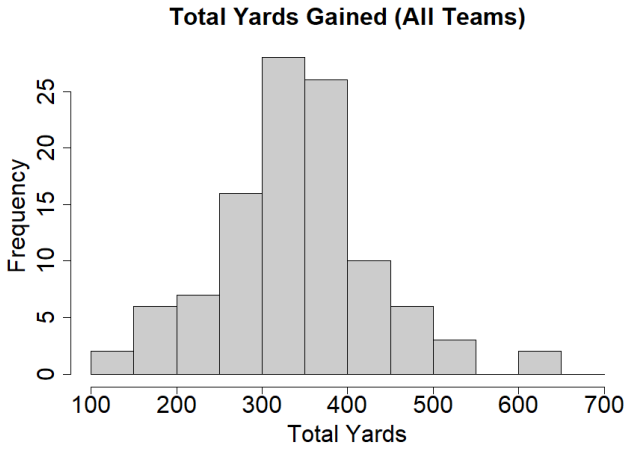
11. **Quantity A:** The mean body weight for male cats
    Quantity B: The median body weight for male cats
    a. Quantity A is greater
    b. Quantity B is greater
    c. The quantities are about the same
    d. The relationship cannot be determined without more information

12. **Quantity A:** The range of body weights for female cats
    Quantity B: The IQR of body weights for male cats
    a. Quantity A is greater
    b. Quantity B is greater
    c. The quantities are about the same
    d. The relationship cannot be determined without more information

---

Use the following information to answer Questions 13-15:
The Super Bowl is the annual championship game of the National Football League (NFL) and is held each year. A football fan collected data about each of the 53 Super Bowl games between 1967 and 2019. The variables *Total Yards* and *Outcome* for each team were recorded in the dataset and are summarized in the plots below.





13. The variable *Total Yards* is a
    a. categorical variable.
    b. quantitative variable.

14. The variable *Outcome* (win, loss) is a
    a. categorical variable.
    b. quantitative variable.

15. The histogram shows the *Total Yards* for all 106 teams in the 53 Super Bowl games. Approximately how many of the teams had between 200 and 300 total yards?
    a. 23
    b. 37
    c. 69
    d. 83

Use the following information to answer Questions 16-18:
Suppose that the distribution of heart rates for medium-sized dogs is normally distributed with mean 115 beats per minute and standard deviation 18 beats per minute (bpm).

16. Approximately what percentage of medium-sized dogs have a heart rate above 97 beats per minute?
    a. 16%
    b. 32%
    c. 50%
    d. 68%
    e. 84%

17. Which of the following is the heart rate for medium-sized dogs at the 84th percentile?
    a. 79 bpm
    b. 97 bpm
    c. 115 bpm
    d. 133 bpm
    e. 151 bpm

18. Approximately 95% of heart rates for medium-sized dogs fall between which of the following values?
    a. 61 and 169 beats per minute
    b. 79 and 151 beats per minute
    c. 97 and 133 beats per minute
    d. 106 and 124 beats per minute

Use the following information to answer Questions 19-21:
A national poll of workplace managers revealed that 47% of managers plan to allow their employees to continue to work remotely in the future, even if it is possible for them to return to the workplace. A local researcher wishes to know if the rate is different for workplaces in Santa Barbara County. The researcher took a random sample of 95 managers from the professional organization and emailed them. Of the 62 managers who replied, 40 (64.5%) reported that they plan to allow their employees to continue to work remotely, and 22 reported that they are unsure or will not allow their employees to continue to work remotely.

19. Which of the following is the correct hypothesized (null) proportion, p_0, we should use to test the researcher's claim?
    a. 0
    b. 0.47
    c. 0.5
    d. 0.645

20. The null hypothesis will be $H_0: p = p_0$, where p_0 is the hypothesized (null) proportion. Which of the following alternative hypotheses should we use to test the researcher's claim?
    a. $H_A: p < p_0$
    b. $H_A: p \neq p_0$
    c. $H_A: p > p_0$

21. The test statistic was calculated to be 2.7634, and the *p*-value for the test was 0.00572. Which of the following conclusions is correct for a test at the $\alpha = 0.05$ significance level?
    a. Since the *p*-value is less than $\alpha = 0.05$, we **fail to reject** the null hypothesis. Our analysis **does not suggest** that the proportion of managers in Santa Barbara County who plan to allow their employees to continue to work remotely in the future differs from the national rate.
    b. Since the *p*-value is less than $\alpha = 0.05$, we reject the null hypothesis. Our analysis suggests that the proportion of managers in Santa Barbara County who plan to allow their employees to continue to work remotely in the future differs from the national rate.
    c. Since the *p*-value is less than $\alpha = 0.05$, we **fail to reject** the null hypothesis. Our analysis **suggests** that the proportion of managers in Santa Barbara County who plan to allow their employees to continue to work remotely in the future differs from the national rate.
    d. Since the *p*-value is less than $\alpha = 0.05$, we **reject** the null hypothesis. Our analysis **does not suggest** that the proportion of managers in Santa Barbara County who plan to allow their employees to continue to work remotely in the future differs from the national rate.

---

22. Which of the following does **NOT** need to be known in order to compute the p-value?
    a. significance level
    b. direction of the alternative hypothesis
    c. value of the test statistic
    d. distribution of the test statistic when the null hypothesis is true

---

Use the following information to answer Questions 23-25.

A national poll of workplace managers revealed that 47% of managers plan to allow their employees to continue to work remotely in the future, even if it is possible for them to return to the workplace. A local researcher wishes to know if the rate is different for workplaces in Santa Barbara County. The researcher took a random sample of 95 managers from the professional organization and emailed them. Of the 62 managers who replied, 40 (64.5%) reported that they plan to allow their employees to continue to work remotely, and 22 reported that they are unsure or will not allow their employees to continue to work remotely.

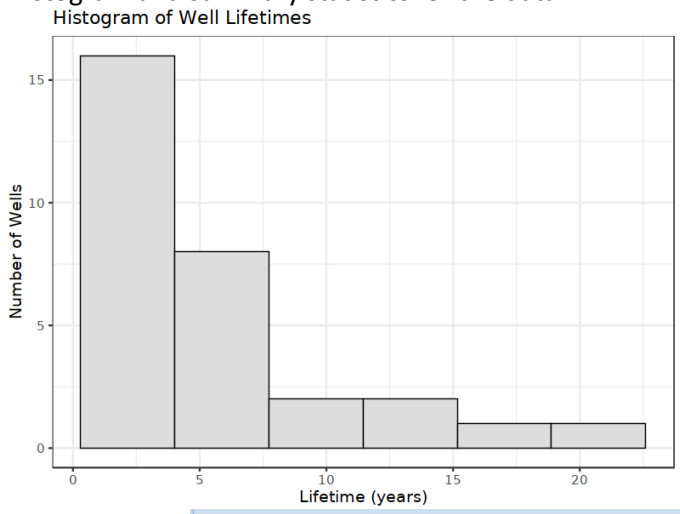
23. Which of the following is the correct hypothesized (null) proportion, p_0, we should use to test the researcher's claim?
    a. 0
    b. 0.47
    c. 0.5
    d. 0.645

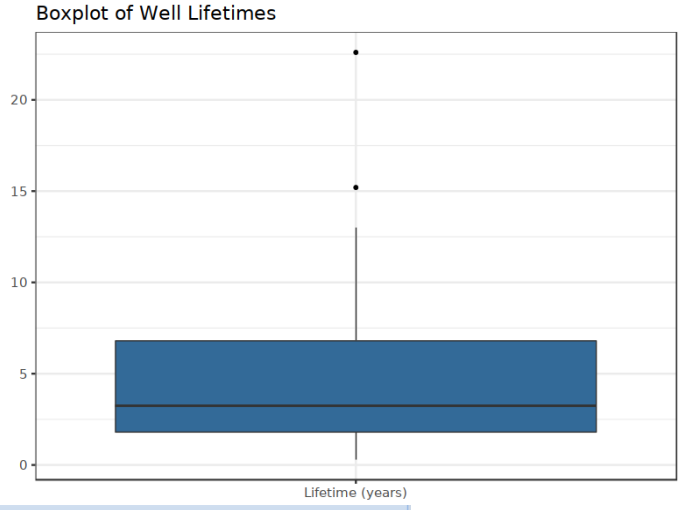
24. The null hypothesis will be $H_0: p = p_0$, where p_0 is the hypothesized (null) proportion. Which of the following alternative hypotheses should we use to test the researcher's claim?
    a. $H_A: p < p_0$
    b. $H_A: p \neq p_0$
    c. $H_A: p > p_0$

25. The test statistic was calculated to be 2.7634, and the *p*-value for the test was 0.00572. Which of the following conclusions is correct for a test at the $\alpha = 0.05$ significance level?
    a. Since the *p*-value is less than $\alpha = 0.05$, we **fail to reject** the null hypothesis. Our analysis **does not suggest** that the proportion of managers in Santa Barbara County who plan to allow their employees to continue to work remotely in the future differs from the national rate.
    b. Since the *p*-value is less than $\alpha = 0.05$, we reject the null hypothesis. Our analysis suggests that the proportion of managers in Santa Barbara County who plan to allow their employees to continue to work remotely in the future differs from the national rate.
    c. Since the *p*-value is less than $\alpha = 0.05$, we **fail to reject** the null hypothesis. Our analysis **suggests** that the proportion of managers in Santa Barbara County who plan to allow their employees to continue to work remotely in the future differs from the national rate.
    d. Since the *p*-value is less than $\alpha = 0.05$, we **reject** the null hypothesis. Our analysis **does not suggest** that the proportion of managers in Santa Barbara County who plan to allow their employees to continue to work remotely in the future differs from the national rate.

*****Free Response Questions*** For the following questions, you would not get full credit unless you show your work. Partial credit is granted based on worked shown.**

26. **Lifetime of Well Pumps** Rural homes often have separate wells for their water supply. In a study on the extra costs of rural living, 30 homes were randomly selected, and the lifetime (in years) of each well pump was recorded. Here are a histogram and summary statistics for the data.





	n	Sample Mean	Standard Deviation	Min	Q1	Median	Q3	Max
Lifetime (years)	30	5.130	5.118	0.3000	1.800	3.250	6.800	22.60

    a. Describe the distribution of well pump lifetimes. Make sure to address all four aspects (recall the SOCS mnemonic).
    The distribution of well pump lifetimes is unimodal and right skewed. The lifetimes are centered at the median of 3.250 years, and the IQR for the well pump lifetimes is Q3 – Q1 = 6.8 – 1.8 = 5.0 years. None of the well pump lifetimes appear to be outliers (the suspected outliers in the boxplot are just part of the long right tail in the distribution).

b. Should you use the mean and standard deviation or the median and IQR to describe the center and spread of the distribution? Clearly state "mean and standard deviation" or "median and IQR" and briefly (in 1-2 sentences) explain.
Since the distribution of well pump lifetimes is skewed, the best measures for center and spread are the median and IQR, respectively. Neither the median nor the IQR is impacted by the few extreme observations in the long right tail of the distribution.

c. Suppose the maximum well pump lifetime was entered as 55.9 instead of 22.6 years. Which would change, the mean, the median, or both? Clearly state "mean," "median," or "both" and briefly (in 1-2 sentences) explain.
The mean would change because it uses the values of all the observations in its calculation. The median would not change because the values in the middle of the data set are unaffected by changes to the values in the extremes.

---

27. **First-Generation College Students** According to the student profile at a large university, 14% of undergraduate students are first-generation college students. Does the proportion of students in a large introductory statistics course who are first-generation college students differ from that of the University? In a random sample of 500 students in the intro stats course, 80 were first-generation college students. We plan to test the following hypotheses:
H_0: The proportion of students in the intro stats course who are first-generation college students is 0.14 ($p = 0.14$).
H_A: The proportion of students in the intro stats course who are first-generation college students differs from 0.14 ($p \neq 0.14$).

a. Calculate the proportion of students in the sample who are first-generation college students. Be sure to include the symbol for this sample statistic.

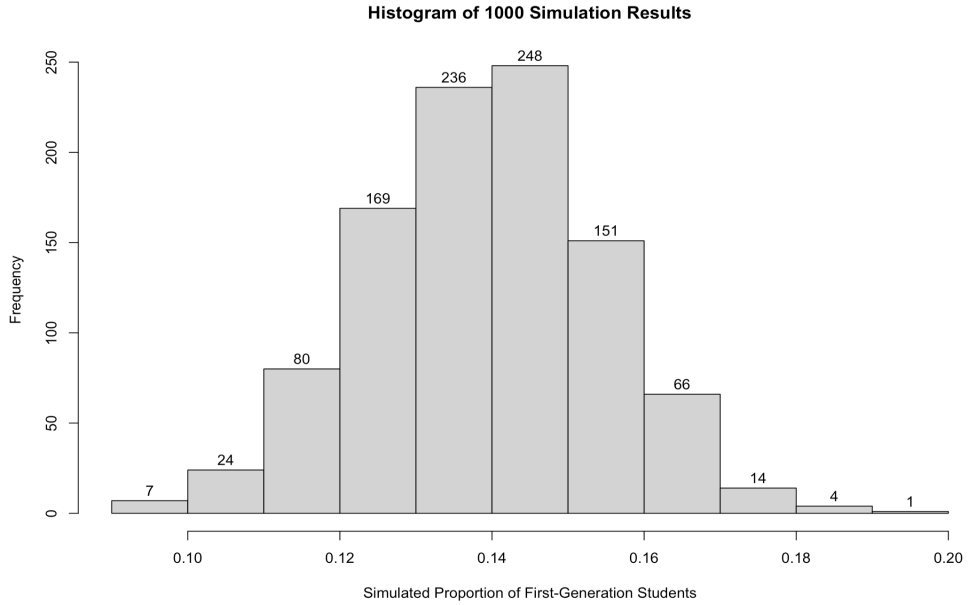
$$\hat{p} = \frac{80}{500} = 0.160$$

b. The paragraph below describes how we might use a simulation-based hypothesis test of the stated hypotheses. We will let blue poker chips represent students who are first-generation college students and yellow poker chips to represent students who are not first-generation college students. We will use a total of 100 poker chips to simulate the null hypothesis model.
   i. How many of these poker chips should be blue?
   14

   ii. To perform the simulation, we thoroughly mix the poker chips, select one from the bag, and record the color before replacing the chip in the bag. How many times should we draw a poker chip in each repetition of the simulation?
   500

c. Assuming the null hypothesis is true, where should the null distribution be centered?
   i. 0
   ii. 0.14
   iii. 0.16
   iv. 0.50

d. The histogram shows the results from 1000 repetitions of the simulation.



Which of the following is true about estimated *p*-value for this hypothesis test?

i. The estimated p-value is less than 0.05
ii. The estimated p-value is between 0.05 and 0.10
iii. The estimated p-value is more than 0.10

e. Do you reject or fail to reject the null hypothesis at the $\alpha = 0.10$ significance level? Why?
Since the p-value is greater than 0.10, we fail to reject the null hypothesis at the $\alpha = 0.10$ significance level.

f. Write a conclusion addressing whether the proportion of first-generation college students in the intro stats class is different from the 0.14 seen at the University level. Be sure to include context (what the question is about) in your conclusion.
The data do not suggest that the proportion of first-generation college students in the intro stats class is different from the 0.14 seen at the University level.